

Board Bring-Up Plan

Raspberry Pi Parcel Defect Detection Camera System · Prepared by Muhammad Hafiz bin Kamaruddin · 11 June 2026

Prerequisites

- **Equipment:** current-limited 12V bench supply, multimeter, oscilloscope, Raspberry Pi 5 power cable/USB-C cable, test camera ribbon, photoelectric sensor, 12V LED bar/ring, relay/PLC test load.
- **Safety:** start with the 12V supply current limit at 500mA before connecting the Raspberry Pi, then increase only after confirming the rails.
- Do not connect the 12V sensor output directly to Raspberry Pi GPIO; verify U4 optocoupler isolation is populated.

1. Visual Inspection

- Confirm D1 Schottky orientation from F1 to +12V_PROTECTED.
- Confirm D2 TVS is across +12V_PROTECTED and GND.
- Confirm C1/C3 electrolytic capacitor polarity and all GND pins are soldered.
- Confirm Q1, Q2, D6, and relay K1 orientation.

2. Power Rail Verification

Rail	Source	Expected Voltage	Measure At	Initial Current Limit	Pass Criteria
+12V_IN	J1	12V nominal	J1 Pin_1 to GND	500mA	No short; supply remains current-limited off
+12V_PROTECTED	F1 + D1	~11.4V to 12V depending D1 drop	D1 K / C1 P1 to GND	500mA	TVS not heating; rail present after fuse/diode
+5.1V_PI	U1 buck output	5.1V target	C3 P1 / J2 VBUS to GND	500mA initially, then >=5A-capable supply path	5.0V to 5.2V under Pi load
+3V3	Raspberry Pi header	3.3V	U2 3V3 pins	Pi powered	3.2V to 3.4V

Procedure:

1. With power off, measure resistance from +12V_PROTECTED to GND and +5.1V_PI to GND.
2. Apply 12V at J1 with the Raspberry Pi disconnected.
3. Verify +12V_PROTECTED and adjust U1 to 5.1V before plugging into the Raspberry Pi.
4. Connect Raspberry Pi 5 only after +5.1V_PI is verified.

3. Critical Signal Verification

Signal	Net Name	Expected State	Measure At	Notes
Sensor trigger	GPIO17_TRIGGER	Pulled high to 3.3V idle; pulled low when U4 conducts	U2 GPIO17	Isolated from 12V SENSOR_OUT by PC817
LED PWM	GPIO18_LED_PWM	0V idle, 3.3V PWM active	R3/U2 GPIO18	Controls Q1 gate through R3
PASS LED	GPIO23_PASS_LED	3.3V when PASS indicator on	R5/U2 GPIO23	Drives D4 through R5
DEFECT LED	GPIO24_DEFECT_LED	3.3V when DEFECT indicator on	R6/U2 GPIO24	Drives D5 red LED through R6
Relay control	GPIO27_RELAY_CTRL	3.3V energizes K1 through Q2	R7/U2 GPIO27	Pi GPIO never drives relay coil directly

4. Connector and Interface Tests

Connector	Function	Pins / Nets	Test Method
J1	12V input	+12V_IN, GND	Verify polarity before power
J2	Raspberry Pi power output	+5.1V_PI, GND	Verify 5.1V before Pi connection
J3	Photoelectric sensor	+12V_PROTECTED, GND, SENSOR_OUT	Trigger sensor and verify GPIO17_TRIGGER toggles
J4	LED bar/ring	+12V_PROTECTED, LED_NEG_SWITCHED	PWM GPIO18 and observe LED brightness
J5	Relay/PLC output	RELAY_COM, RELAY_NO, RELAY_NC_LABEL	Toggle GPIO27 and verify contact continuity
U3	CSI camera representation	CSI ribbon cable/module	Connect Raspberry Pi Camera Module 3 Wide to Pi CSI port, not GPIO

5. Raspberry Pi / Software Bring-Up

- Boot Raspberry Pi 5 from known-good OS image.
- Enable camera interface and verify Camera Module 3 Wide capture.
- Configure GPIO17 as input with external pull-up via R2.
- Configure GPIO18 as PWM output.
- Configure GPIO23, GPIO24, and GPIO27 as digital outputs.

6. Functional Validation

Test	Input	Expected Output	Pass Criteria
Sensor trigger	Block photoelectric sensor beam	GPIO17_TRIGGER changes state	Pi software detects parcel arrival
LED lighting	PWM GPIO18	J4 LED bar brightness changes	Smooth dimming, Q1 remains cool
PASS indicator	Set GPIO23 high	D4 green LED on	Visible PASS indication
DEFECT indicator	Set GPIO24 high	D5 red LED on	Visible DEFECT indication
Relay/PLC/ejector	Set GPIO27 high	K1 energizes, J5 contact changes	GPIO isolated from coil by Q2 and D6 clamps flyback
Camera	Capture image over CSI	Image frame available	Parcel image captured without GPIO camera wiring

Notes / Open Items

- U1 is represented as a buck module; the final hardware must use a 12V-to-5.1V converter rated at minimum 5A for Raspberry Pi 5.
- K1 library part is SPST-NO; J5 includes an NC-labelled terminal for proposal completeness, but an actual SPDT relay should be selected if NC contact is required in production.
- U3 is a schematic representation of the CSI camera/ribbon connection; the camera connects to the Raspberry Pi CSI port, not to the GPIO header.

Pass/Fail Summary

Section	Result	Notes
Visual Inspection	☐ Pass / ☐ Fail	
Power Rails	☐ Pass / ☐ Fail	
Critical Signals	☐ Pass / ☐ Fail	
Connectors	☐ Pass / ☐ Fail	
Raspberry Pi Software	☐ Pass / ☐ Fail	
Functional Validation	☐ Pass / ☐ Fail	

Board Status: ☐ Ready for prototype test ☐ Requires rework

Tested by: _____ Date: _____ Signature: _____